DemocraticNet: Your shield against misinformation!

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Abstract. The proliferation of fake news on social media platforms has become a critical concern, undermining the integrity of democratic processes and eroding public trust in governmental institutions. Misinformation spreads rapidly through these channels, complicating the public's ability to discern fact from fiction and exacerbating political polarization. In this paper, we present DemocraticNet: Your Shield Against Misinformation! a state-of-the-art system designed to classify news articles by the relevant government ministry using advanced natural language processing and deep learning techniques. Our approach leverages cutting-edge models, including BERT, DebertaV2, RoBERTa, XLNet, and XLM-RoBERTa, to automatically categorize news content and identify potentially misleading information. By forwarding classified news to the appropriate ministries for further analysis, our system provides a proactive mechanism for mitigating the spread of misinformation. Through rigorous experimentation, we evaluate the efficacy of various model-optimizer combinations to ensure robust and accurate classification. DemocraticNet aims to enhance the resilience of democratic nations against the pervasive threat of digital misinformation, fostering informed civic participation and preserving the integrity of elections and policy debates.

Keywords: Deep Learning, News Classification, Misinformation, Electoral Integrity

1 Introduction

The proliferation of fake news, particularly on social media platforms, has become a significant concern due to its detrimental effects on governmental institutions and democratic processes. Fake news, characterized by intentionally false or misleading content presented as real news, has leveraged the rapid dissemination capabilities of social media to spread misinformation to large audiences. This phenomenon has made it increasingly difficult for citizens to discern truth from fiction, introducing confusion and polarization into political discourse and potentially undermining the functioning of democracy.

Experts warn that the unchecked spread of fake news could decrease public trust in societal institutions and make elections more vulnerable to external influence through targeted misinformation campaigns. In response, governments worldwide have considered various solutions, including social media regulations, public education programs, and third-party fact-checking initiatives. However, identifying an appropriate and ethical approach remains challenging, given the need to balance efforts against misinformation with concerns over limiting freedom of speech.

This paper presents a state-of-the-art method for classifying news articles according to their relevance to specific government ministries. The proposed system leverages advanced natural language processing, machine learning techniques, and rule-based systems to automatically categorize news content and flag potentially misleading information for further scrutiny by relevant government authorities. By implementing such a system, the goal is to provide a proactive mechanism for identifying and mitigating the spread of misinformation within a country's information ecosystem.

This approach aims to empower government ministries to stay informed about news narratives relevant to their respective domains and take necessary actions to counter the dissemination of fake news. Ultimately, this study seeks to contribute to broader efforts aimed at preserving the integrity of democratic processes and upholding the principles of an informed citizenry by leveraging cutting-edge technological solutions to combat the growing challenge of fake news propagation.

2 Background

In the modern digital age, the proliferation of fake news has emerged as a significant threat to the legitimacy and effectiveness of governments worldwide. This insidious web of misinformation and disinformation has woven a complex tangle, casting a long shadow over public trust in elected officials and government institutions. The corrosive nature of fake news poses a multifaceted challenge that demands a coordinated and comprehensive response from various stakeholders.

The detrimental impact of fake news on governments manifests itself in several concerning ways:

1. Erosion of Public Trust: Fabricated stories about corruption, scandals, or policy failures can sow seeds of doubt and suspicion among the populace when widely spread. Imagine headlines falsely accusing government officials of secretly plotting against their own citizens. Such narratives, even if demonstrably false, can damage public perception and make genuine government initiatives seem tainted, undermining the government's credibility.
2. Hindered Decision-Making: Policymakers rely heavily on accurate data and public feedback when navigating complex issues such as healthcare, economic reform, or national security. However, when the information

landscape is polluted with misinformation, making informed decisions becomes a herculean task. Fake news can skew both data and public sentiment, leading to policies based on false premises and public backlash towards decisions with legitimate reasoning, obscured by a fog of fabricated narratives.

3. Undermining Democratic Processes: In democratic societies, free and fair elections are the cornerstone of governance. The unchecked spread of fake news can make elections vulnerable to external influence via targeted misinformation campaigns aimed at swaying the choices of specific groups of voters. This threat to the integrity of electoral processes strikes at the heart of democratic principles and the legitimacy of elected officials.

Addressing the fake news challenge is a complex battle with no easy solutions. Balancing the need to curb harmful misinformation with the fundamental right to freedom of speech and expression is a delicate dance. Finding effective solutions requires collaboration between governments, technology companies, media organizations, and citizens themselves, as well as a multi-pronged approach that leverages various technological, legal, and educational initiatives.

In this context, advances in machine learning and deep learning have opened up new possibilities for automatically categorizing news content by the government ministry or ministries it is most relevant to. By feeding these artificial intelligence (AI) systems large datasets of news articles that have already been manually assigned ministry tags, they can learn to make these categorizations on their own with a high degree of accuracy. For example, a news article discussing changes in educational policy would be tagged as belonging to the Ministry of Education, while one about infrastructure spending may be designated for both the transportation and construction ministries.

Powerful neural network architectures could process an incoming news article and almost instantly predict the most probable ministry or ministries it should be routed to. This automated categorization could occur behind the scenes, ensuring relevant officials and analysts are kept informed of developments concerning their areas of governance. As more labeled data is provided, the machine learning models can grow even more sophisticated at multi-label classification, developing subtler text understandings and modeling interdependencies between ministries.

These AI systems have the potential to become versatile automated assistants, relieving manual categorization burdens in government newsrooms and

communication teams. Their integration promises to usher news filtering and dissemination into the era of AI-augmented governance, enabling more targeted and responsive flows of information across all levels of government.

The rapid advances in content-based predictive modeling hold immense potential for strengthening evidence-based policymaking and decision-making. By harnessing AI to accurately categorize and route news content to the appropriate ministries and agencies, policymakers can stay better informed about developments relevant to their domains, fostering a more robust connection between policies and accurate, timely data.

In the face of the fake news onslaught, governments must embrace innovative technological solutions that can help them navigate the deluge of information, separating fact from fiction and ensuring credible news reaches the right decision-makers. Only by combining such cutting-edge approaches with broader efforts to promote media literacy, strengthen fact-checking mechanisms, and uphold the principles of a free and responsible press can societies hope to mitigate the corrosive effects of fake news and preserve the integrity of their democratic institutions.

3 Introduction to text classification method

Text classification is a fundamental technique in natural language processing (NLP) that automatically categorizes text documents into predefined classes. This process involves training a machine learning model or neural network on a labeled dataset to recognize patterns and features, which it then uses to predict categories for new, unseen text data.

Traditionally, text classification relied on machine learning algorithms such as Support Vector Machines (SVM), Naive Bayes, or Decision Trees. These models used handcrafted features like bag-of-words or TF-IDF vectors to represent text data. While these approaches achieved reasonable accuracy, they often struggled to capture complex linguistic patterns and semantic relationships in the text.

The advent of deep learning revolutionized text classification by enabling models to automatically learn hierarchical representations of text data. Deep learning models, such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), or Transformers, can learn feature representations directly from raw text inputs. These models have shown remarkable performance improvements over traditional approaches, capturing intricate linguistic patterns, semantic relationships, and contextual information, resulting in more accurate and robust text classification.

Multi-label text classification extends this concept by allowing documents to belong to multiple categories simultaneously. Instead of assigning a single label, the model predicts a set of labels, each representing a different category. This approach is particularly useful for text data with overlapping themes. Deep learning-based models like multi-label CNNs or Transformers have been successfully applied to these tasks, leveraging their ability to learn complex representations and capture dependencies between different labels.

Recent advancements in text classification have been driven by novel deep learning techniques. Researchers have explored architectures such as hierarchical models, attention mechanisms, and pre-trained language models like BERT, GPT, and RoBERTa. These models have achieved state-of-the-art results on various text classification benchmarks and have become the current trend in research and industry applications.

The shift from traditional machine learning to deep learning has significantly improved text classification performance and opened up new possibilities for solving complex NLP tasks. By leveraging these advanced models, we can achieve more accurate and robust text classification results, enabling applications in various domains such as news classification, sentiment analysis, and document organization.

4 Related Work

To establish a comprehensive understanding of state-of-the-art methods, an extensive literature review was conducted. This involved a thorough analysis of various techniques implemented by authors in the field of news-related text classification. The resulting Table 1 presents a valuable resource, encompassing the title, authors, year, and a concise description of each method. This compilation serves as a reference point, enabling the identification of the most accurate and suitable approach for the current research task. The literature review process entailed a meticulous selection of relevant papers and the extraction of key information, ensuring a diverse range of approaches and insights were gathered. This well-rounded analysis provides authors and practitioners with a comprehensive overview of the latest advancements in text classification.

Table 1. State-of-the-Art Deep Learning Models for News Classification

Title	Authors	Year	Brief Description
Fake News Classification Based on Content Level Features [6]	Lia et. al.	2022	Researchers utilized ML and NLP techniques to develop a system for identifying fake news on Twitter. Their approach, which employed neural network models trained on a large, labeled dataset from Twitter,

			demonstrated superior performance compared to traditional ML models. On average, the neural network models achieved approximately 6% higher precision scores, with some models reaching up to 90% accuracy in fake news classification.
Automated Amharic News Categorization Using Deep Learning Models [7]	Endalie et. al.	2021	The researchers employed fastText, a library for efficient text classification and representation, to generate vector embeddings that capture the semantic meaning of texts. This approach addressed the limitations of traditional methods. The resulting text vector matrix was then input into the embedding layer of a CNN architecture. The CNN automatically extracted relevant features from the text embeddings. Ultimately, this model achieved an impressive 93.79% accuracy in classifying the text data.
BerConvoNet: A deep learning framework for fake news classification [2]	Choudhary et. al.	2021	The researchers introduced BerConvoNet, a deep-learning framework designed to classify news articles as either real or fake. BerConvoNet comprises two primary components: a news embedding block (NEB) and a multi-scale feature block (MSFB). The NEB is responsible for generating embeddings that represent the textual content of the news articles, while the MSFB extracts features at multiple scales from these embeddings, enabling the model to capture both local and global patterns in the text data.
Different Machine Learning-based Approaches of Baseline and Deep	Rabib Hossain et. al.	2020	This study compares various machine learning techniques for categorizing Bengali news articles. The approaches include traditional models like Naive Bayes and Logistic

Learning Models for Bengali News Categorization [5]			Regression, alongside advanced deep learning models like CNN and Bi-LSTM.
Research on hot news classification algorithm based on deep learning [3]	Wang et. al.	2019	This researcher investigates the application of deep learning for text categorization, specifically focusing on news text. The proposed model leverages a double Bi-Directional Gated Recurrent Unit (Bi-GRU) with an attention mechanism to predict news hotspots and achieves promising results.

To identify relevant news classification datasets for this study, an extensive literature review was undertaken. The primary objective of this review was to gather comprehensive information about the diverse datasets employed by researchers in the field of news classification. The resulting compilation presents a valuable resource in the form of Table 2, which includes the dataset name accompanied by a concise description of each dataset's characteristics. This structured overview serves as a valuable reference point, enabling the identification of the most pertinent and appropriate dataset for the current research endeavor.

Table 2. Overview of Available Datasets in the Domain of News Classification

Dataset	Brief Description
Reuters News [14]	The dataset used is multi-class and multi-label, containing 90 distinct categories. It includes 7769 documents for training and 3019 for testing purposes.
Sogou News [13]	The Sogou News dataset combines articles from the SogouCA and SogouCS collections. News articles are categorized based on the domain name found within their URL.
AG News [12]	Compiled by the academic search engine ComeToMyHead, the AG News dataset contains news articles gathered from over 2,000 sources. It offers 120,000 samples for training and 7,600 for testing. Each sample consists of a brief news item categorized into one of four classes.
Newsgroups [15]	Compiled by the academic search engine ComeToMyHead, the AG News dataset contains news articles gathered from over 2,000 sources. It offers 120,000 samples for training and 7,600 for testing. Each sample consists of a brief news item categorized into one of four

	classes.

In addition to the datasets identified through the literature review, several other noteworthy datasets have been developed specifically for news categorization tasks. These include the BBC News Dataset [18], the Google News Dataset [17], and the Bing News Dataset [16]. These datasets have been widely utilized by researchers and practitioners in the field, providing valuable resources for training and evaluating news classification models.

5 Dataset Creation

As part of our objective to classify news articles according to the ministries of the Indian government, we encountered a significant lack of available datasets tailored to this specific task. To address this gap, we took the proactive initiative to create our own comprehensive dataset specifically designed for this research endeavor.

The construction of this dataset involved a meticulous process of merging three existing news datasets sourced from the Kaggle platform. Subsequently, we undertook the labor-intensive task of manually labeling each news article according to its relevance and alignment with the corresponding ministries of the Indian government. It is important to note that while the Indian government comprises over 100 ministries, our dataset was constrained to consider 59 ministries due to limitations in data availability and other practical constraints. To account for news articles that may not fall under the purview of any specific ministry, we introduced an additional label, bringing the total number of labels to 60, with the 60th label representing the "no ministry" category.

Recognizing the value of open access and collaboration in advancing research, we have made this meticulously curated dataset openly accessible under the permissive MIT license. The dataset can be conveniently downloaded from the following Kaggle link: https://www.kaggle.com/neelshah2022/code. However, we kindly request that this dataset be utilized ethically and in a manner that contributes to the betterment of society.

The dataset itself is a substantial compilation, comprising 36,051 news articles spanning across 60 distinct classification classes or labels. It is important to highlight that there are no duplicate news articles within the dataset, ensuring its integrity and reliability for training and evaluation purposes. For the sake of clarity and ease of reference, we have provided a comprehensive list of the ministries and their corresponding label numbers that were utilized in the creation of this dataset and they are shown in Table 3.

Table 3. Ministries and Their Corresponding Labels in the Dataset

Label	Ministry
0	Ministry of Railways

1	Ministry of Rural Development
2	Ministry of Steel
3	Ministry of Science & Technology
4	Ministry of Information & Broadcasting
5	Ministry of Food Processing Industries
6	Ministry of Health and Family Welfare
7	Ministry of Human Resource Development
8	Ministry of Agriculture
9	Ministry of Environment and Forests
10	Ministry of Chemicals and Fertilizers
11	Ministry of Water Resources
12	Ministry of Defence
13	Ministry of Petroleum & Natural Gas
14	President's Secretariat
15	Ministry of Micro, Small & Medium Enterprises
16	Ministry of Mines
17	Ministry of Tourism
18	Ministry of Housing & Urban Affairs
19	Ministry of Coal
20	Prime Minister's Office
21	Ministry of Textiles
22	Ministry of Commerce & Industry
23	Ministry of External Affairs
24	Ministry of Social Justice & Empowerment
25	Ministry of Power

26	Ministry of Consumer Affairs, Food & Public Distribution
27	Ministry of Heavy Industries & Public Enterprises
28	Ministry of Communications
29	Ministry of Shipping
30	Ministry of Finance
31	Ministry of Tribal Affairs
32	Ministry of Statistics & Programme Implementation
33	Ministry of Labour & Employment
34	Ministry of Law & Justice
35	Vice President's Secretariat
36	Ministry of Civil Aviation
37	Ministry for Development of North-East Region
38	UPSC Department
39	Ministry of Agro & Rural Industries
40	Ministry of Home Affairs
41	Ministry of Youth Affairs and Sports
42	Special Service and Features
43	Ministry of New and Renewable Energy
44	Ministry of Parliamentary Affairs
45	Planning Commission
46	Ministry of Personnel, Public Grievances & Pensions
47	Election Commission
48	Department of Space
49	Ministry of Disinvestment
50	Department of Ocean Development

51	Ministry of Overseas Indian Affairs
52	Ministry of Housing and Urban Poverty Alleviation
53	Ministry of Culture
54	Ministry of Company Affairs
55	Ministry of Panchayati Raj
56	Cabinet Committee on Economic Affairs (CCEA)
57	Cabinet
58	Department of Atomic Energy
59	Cabinet Committee Decisions
60	No Ministry

Through this painstaking effort, we have created a valuable resource that addresses a significant gap in the availability of datasets tailored to the classification of news articles according to the ministries of the Indian government. We hope that this dataset will prove invaluable to researchers, practitioners, and stakeholders alike, fostering advancements in the field and contributing to the broader goal of information organization and accessibility.

Table 4. provides sample examples from the dataset, including the news title, news content, and the corresponding ministry associated with each news article. This serves as a demonstration of the dataset's structure and content.

News Id	News Title	News Content	Ministries Associated
24226	McCullum messaged me asking if I was in Moeen on Test Cricket comeback	England all-rounder Moeen Ali, who announced retirement from Test cricket last year, has revealed that England's newly-appointed Test coach Brendon McCullum has asked him if he would be available to play Tests again in the future. "We spoke...[he asked] Would I be available [in the future]? I said, 'Call me at the time'. We'll see...The door is open," stated	Ministry of Youth Affairs and Sports (41)

		Moeen.	
24263	2 laborers die as soil collapses at construction site in Gurugram	At least two laborers died after soil collapsed on them at an under-construction plot in Gurugram's Sector 57 on Wednesday, police said. "The basement had been dug over 10 feet below the ground level... laborers were putting pillars...when a mound of soil collapsed on them...they were trapped underneath," police added. It took over 15 minutes to retrieve their bodies, police stated.	Ministry of Health (7) and Family Welfare, Ministry of Labour & Employment (33)

6 Statistics of our dataset

Our original dataset comprised approximately 36,000 news articles, each classified into one of 59 different ministry categories. However, preliminary analysis shown in Figure 1 with the help of the bar graph revealed a significant class imbalance, with only 299 articles corresponding to 20 specific ministry classes. This imbalance poses a substantial challenge for training robust deep learning models, as models trained on such data tend to be biased towards the more frequently occurring classes, leading to poor generalization of the underrepresented classes.

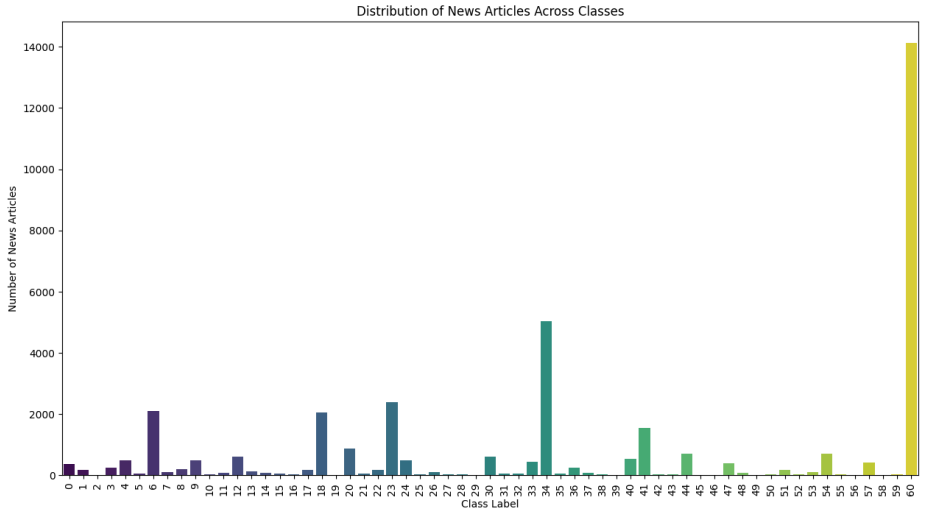


Figure 1: Distribution of News Across Different Ministries (Classes), with -1 Indicating "No Ministry"

To address this issue, we implemented a data preprocessing step aimed at balancing the dataset. Specifically, we removed the 299 articles associated with the 20 underrepresented ministry classes. This exclusion process resulted in a more balanced dataset, reducing the number of ministry classes from 59 to 39. Although there was also some imbalance in the remaining part of the dataset, it was acceptable because the number of news articles per class was sufficiently large.

Specifically, the ministries that were excluded during the preprocessing were (2) Ministry of Steel, (10) Ministry of Chemicals and Fertilizers, (16) Ministry of Mines, (19) Ministry of Coal, (25) Ministry of Power, (27) Ministry of Heavy Industries & Public Enterprises, (28) Ministry of Communications, (29) Ministry of Shipping, (38) UPSC Department, (39) Ministry of Agro & Rural Industries, (43) Ministry of New and Renewable Energy, (45) Planning Commission, (46) Ministry of Personnel, Public Grievances & Pensions, (49) Ministry of Disinvestment, (50) Department of Ocean Development, (52) Ministry of Housing and Urban Poverty Alleviation, (55) Ministry of Panchayati Raj, (56) Cabinet Committee on Economic Affairs (CCEA), (58) Department of Atomic Energy, and (59) Cabinet Committee Decisions.

Additionally, we introduced a new category labeled "No Ministry" to account for articles that did not fit into any of the remaining ministry categories, bringing the total number of classes to 40. The steps taken in this preprocessing phase are essential for mitigating the risks associated with imbalanced datasets. They help ensure that the trained model can provide accurate and unbiased predictions, which is critical for the integrity and utility of our research in the field of AI and its applications in news categorization and ministry classification.

7 Methodology

In the ever-evolving landscape of natural language processing, a new breed of language models has emerged, each with its unique strengths and capabilities. Among these marvels of modern technology, BERT [8] stands tall as the pioneer, a bidirectional transformer that revolutionized the way we approach language understanding. Developed by the brilliant minds at Google, BERT's ability to capture context from both directions set a new standard for NLP tasks, including text classification.

However, as with any groundbreaking innovation, there was room for improvement. Enter DebertaV2 [9], a variant of BERT that introduces disentangled attention and enhanced decoding, further refining the model's performance across a wide range of tasks. Like a skilled artisan honing a masterpiece, DebertaV2 built upon BERT's foundation, pushing the boundaries of what was thought possible.

Not to be outdone, researchers at Facebook unveiled RoBERTa [10], a robustly optimized version of BERT that employed a different pretraining approach and a larger dataset. With its enhanced capabilities, RoBERTa quickly established itself as a formidable contender in the realm of text classification, outperforming its predecessor in various benchmarks.

Meanwhile, a different approach emerged in the form of XLNet [11], an autoregressive language model that captured bidirectional context in a unique way. Unlike BERT, which processed input sequences simultaneously, XLNet adopted a sequential approach, allowing it to excel in tasks that required a deeper understanding of language structure and dependencies.

Lastly, we have XLM-RoBERTa [22], a multilingual variant of RoBERTa that transcends language barriers. Pretrained on a vast corpus of text data from multiple languages, XLM-RoBERTa brings the power of RoBERTa to a global stage, enabling text classification and other NLP tasks to be performed seamlessly across linguistic boundaries.

In this diverse ecosystem of language models, each approach offers distinct advantages and Trade-offs. While BERT and its variants excel in capturing contextual information, XLNet's sequential approach may be better suited for tasks that require a more nuanced understanding of language structure. Similarly, XLM-RoBERTa's multilingual capabilities make it an invaluable asset for organizations operating in a globalized world.

When it comes to classifying documents for government ministries, the choice of language model depends on the specific requirements of the task at hand. For example, if the classification involves a diverse set of languages, XLM-RoBERTa may be the preferred choice. On the other hand, if the focus is on capturing fine-grained contextual nuances within a single language, DebertaV2 or RoBERTa could be more appropriate.

The key to successful text classification lies in understanding the strengths and limitations of each model, and carefully evaluating the Trade-offs between performance, computational resources, and the unique characteristics of the data being processed.

8 Experimentation and Results

To determine the most suitable model for our multiclass news classification task, we conducted a series of experiments with different combinations of language models and optimizers. This systematic evaluation allowed us to assess the strengths and trade-offs of each approach, ensuring that the chosen model effectively captures the contextual nuances and linguistic intricacies present in the news articles spanning various domains and ministries.

Different experiments were performed with the combinations of BERT, DebertaV2, RoBERTa, and XLNet models, coupled with optimizers like SGD [23], Adam [1], AdamW [26], and NAdam [4]. The performance of each combination was meticulously evaluated using top-3 accuracy and loss percentages, on a held-out validation set.

Additionally, we experimented with fine-tuning the models for around 100 epochs, during which we observed that the accuracy started declining, and the loss began to increase after a certain point. Interestingly, we found that the models achieved their highest top-3 accuracy and the lowest loss within the first 15 epochs when fine-tuned with optimizers like AdamW, NAdam, and Adam. Beyond this point, the models' performance started to deteriorate. However, this trend was observed for all optimizers except SGD, where the results were not as promising. With SGD, the top-3 accuracy remained quite low, and the loss continued declining with every additional epoch, failing to converge within the specified range.

To ensure that we captured the best results, we employed the callback functionality to track the models' performance during fine-tuning. Consequently, for the results presented in Table 5, we have considered the best accuracy and loss values obtained during the fine-tuning process, which occurred within the initial 15 epochs for optimizers like AdamW, but we had to consider the final epoch results for SGD due to its divergent behavior.

Through this rigorous experimental process, we gained valuable insights into the effectiveness of each language model and optimizer combination for our specific multiclass news classification task. The findings from these experiments guided our decision-making process, enabling us to select the most appropriate approach that strikes the right balance between accuracy, computational efficiency, and practical considerations.

For the experiments, the models were compiled utilizing BCEWithLogitsLoss as the loss function. During the fine-tuning process, a consistent learning rate of 2e-5 and a weight decay of 1e-6 were employed across all model-optimizer combinations. The maximum text length, including the title and the text in the news articles, was capped at 120 words. We have considered 80 percent of the dataset for training and 20 percent for validation. To leverage the computational power of multiple GPUs and accelerate training, we employed MirroredStrategy, enabling parallel training across available hardware resources. All models were fine-tuned using multiple NVIDIA RTX A5000 and NVIDIA T4 GPUs with a batch size of 64.

The evaluation of model performance was based on Top-3 accuracies and loss values, which served as reliable indicators for predicting the most appropriate class assignments. By analyzing these metrics, we could make informed decisions about which class the model should predict, given the input data. This approach allowed us

to leverage the strengths of each model-optimizer combination and optimize the classification process for maximum accuracy and efficiency.

Table 5. Experimental Results on the Newly Labeled News Dataset

Model Name	Optimizer	Top-3 Accuracy	Loss
microsoft/deberta-base	SGD	58.700%	13.458%
microsoft/deberta-base	Adam	82.809%	6.593%
microsoft/deberta-base	Nadam	82.683%	6.685%
microsoft/deberta-base	AdamW	78.229%	9.752%
FacebookAI/roberta-base	SGD	07.608%	7.603%
FacebookAI/roberta-base	Adam	81.915%	7.077%
FacebookAI/roberta-base	NAdam	81.273%	7.878%
FacebookAI/roberta-base	AdamW	82.544%	6.551%
xlnet-base-cased	SGD	58.385%	15.078%
xlnet-base-cased	Adam	79.974%	9.806%
xlnet-base-cased	Nadam	84.583%	6.282%
xlnet-base-cased	AdamW	79.318%	9.945%
google-bert/bert-base-uncased	SGD	29.842%	76.227%
google-bert/bert-base-uncased	Adam	77.810%	9.672%
google-bert/bert-base-	Nadam	81.650%	7.593%

uncased			
google-bert/bert-base-uncased	AdamW	79.220%	9.334%
xlm-roberta-base	SGD	02.681%	7.7984%
xlm-roberta-base	Adam	81.985%	7.798%
xlm-roberta-base	Nadam	85.043%	6.063%
xlm-roberta-base	AdamW	81.985%	7.798%

The extensive experimental results presented in the table highlight the intricate interplay between model architectures and optimization techniques in the context of multiclass, multi-label news classification for government ministries. Among the models evaluated, the deberta-base consistently demonstrated robust performance, achieving high accuracy and low loss values across various optimizers.

Notably, when fine-tuned with the Adam and Nadam optimizers, the deberta-base model outperformed most other model-optimizer combinations, attaining top-3 accuracy scores of 82.8% and 82.7%, respectively. This suggests that the deberta-base architecture, coupled with adaptive learning rate optimization techniques like Adam and Nadam, is well-suited for this complex classification task, effectively capturing the nuances of textual data and adapting to the multi-label nature of news articles.

The FacebookAI/roberta-base model also exhibited promising results, particularly when paired with the AdamWoptimizer. It secured the third-highest accuracy of 82.5% while maintaining a relatively low loss of 6.55%. This highlights the efficacy of the roberta-base architecture and the potential benefits of weight decay regularization employed by the AdamW optimizer in preventing overfitting and improving generalization performance.

Interestingly, the xlnet-base-cased model demonstrated contrasting performance depending on the optimizer used. When trained with the AdamW optimizer, it achieved a reasonably high accuracy of 79.3%. However, its performance significantly deteriorated when trained with the SGD optimizer, resulting in an accuracy of only 58.4% and a substantially higher loss of 15.07%. This stark difference underscores the importance of carefully selecting the appropriate optimizer for a given model architecture to ensure optimal convergence and generalization, as the choice of optimizer can significantly impact the model's ability to effectively capture the underlying patterns in the data. It's worth noting that the SGD optimizer consistently underperformed compared to adaptive methods like Adam and its variants. This aligns with the general trend in deep learning, where adaptive optimizers have proven more effective in navigating complex loss landscapes and converging to better solutions.

The google-bert/bert-base-unc-cased and xlm-roberta-base models exhibited varying degrees of success across different optimizers, with some combinations yielding relatively high losses or missing data points. While these models have shown

strong performance in other natural language processing tasks, their suitability for this specific multiclass, multi-label classification task may require further investigation and fine-tuning to unlock their full potential.

Overall, the results underscore the importance of carefully considering the interplay between model architectures and optimization techniques in the context of complex text classification tasks. The deberta-base model, combined with adaptive learning rate optimizers like Adam and Nadam, emerged as a promising candidate for accurately classifying news articles into relevant government ministries, potentially enabling more effective information dissemination and decision-making processes within governmental institutions.

9 Conclusion

Our model leverages advanced natural language processing and machine learning techniques to provide ministries with a robust solution for combating misinformation. Through personalized news content classification, it enables ministries to effectively identify and mitigate the spread of fake news, safeguarding the public from harmful misinformation.

Furthermore, our model offers data-driven insights that inform policy-making and decision-making processes, promoting evidence-based governance. It facilitates inter-ministerial collaboration by enabling the sharing of news and insights, fostering a coordinated approach to complex issues.

During emergencies, our model ensures rapid identification and dissemination of critical news, mitigating the spread of misinformation and panic. Additionally, it provides valuable insights into public sentiment and media coverage, allowing ministries to tailor their communication strategies for effective audience engagement.

By harnessing the power of advanced NLP and ML techniques, our model empowers ministries with a comprehensive solution for combating misinformation, informing decision-making, enabling collaboration, managing crises, and shaping communication strategies, ultimately contributing to more effective and transparent governance.

10 Future Scope

- **Dataset Expansion:** The current model utilizes a dataset of approximately 36,000 news articles. Future work should focus on leveraging a significantly larger and more diverse dataset. This expansion would enable the model to learn more intricate patterns and nuances, thereby enhancing its accuracy and generalization capabilities.
- **Comprehensive Ministry Coverage**: Presently, the model is tailored for 49 ministries of the Indian government. Future efforts should aim to incorporate all existing ministries within the country. This broader scope would provide a more

holistic and inclusive approach to news classification and analysis for the Indian government.

- **International Adaptation:** While developed for the Indian context, the underlying principles and techniques can be extended to other democratic nations. Countries such as the United States, the United Kingdom, and Germany could benefit from adapting similar models to their governmental structures. This cross-cultural application could foster international collaborations and contribute to global efforts in combating misinformation.
- **Societal Impact Evaluation:** A comprehensive assessment of the model's societal impact in India would be valuable. This analysis would provide insights into the advantages and potential challenges of its deployment, guiding policymakers and stakeholders in decision-making. Additionally, it could serve as a blueprint for other nations considering similar implementations.
- **Multimodal Data Integration:** Future research could explore incorporating multimodal data sources, such as images, videos, and audio, into the model's architecture. This integration would enable a more holistic understanding of news content and potentially enhance the model's ability to detect and counter misinformation across multiple channels.

References

1. Xie, Xingyu, et al. "Adan: Adaptive nesterov momentum algorithm for faster optimizing deep models." arXiv preprint arXiv:2208.06677 (2022).
2. Choudhary, M., Chouhan, S. S., Pilli, E. S., & Vipparthi, S. K. (2021). BerConvoNet: A deep learning framework for fake news classification. Applied Soft Computing, 110, 107614.
3. Research on hot news classification algorithm based on deep learning Zaiying Wang, Bohao Song Xi'an University of Science & Technology, Xi'an, Shaanxi, 710054, China
4. Loshchilov, Ilya, and Frank Hutter. "Decoupled weight decay regularization." arXiv preprint arXiv:1711.05101 (2017).
5. Lai, C. M., Chen, M. H., Kristiani, E., Verma, V. K., & Yang, C. T. (2022). Fake news classification based on content level features. Applied Sciences, 12(3), 1116.
6. Endalie, D., & Haile, G. (2021). Automated Amharic news categorization using deep learning models. Computational Intelligence and Neuroscience, 2021.
7. Devlin, Jacob, et al. "Bert: Pre-training of deep bidirectional transformers for language understanding." arXiv preprint arXiv:1810.04805 (2018).
8. He, Pengcheng, et al. "Deberta: Decoding-enhanced bert with disentangled attention." arXiv preprint arXiv:2006.03654 (2020).
9. Liu, Yinhan, et al. "Roberta: A robustly optimized bert pretraining approach." arXiv preprint arXiv:1907.11692 (2019).
10. Yang, Zhilin, et al. "Xlnet: Generalized autoregressive pretraining for language understanding." Advances in neural information processing systems 32 (2019).
11. Zhang, X., Zhao, J., & LeCun, Y. (2015). Character-level convolutional networks for text classification. Advances in neural information processing systems, 28.
12. Sun, C., Qiu, X., Xu, Y., & Huang, X. (2019). How to fine-tune bert for text classification?. In Chinese Computational Linguistics: 18th China National Conference, CCL 2019, Kunming, China, October 18–20, 2019, Proceedings 18 (pp. 194-206). Springer International Publishing.
13. Reuters. [n.d.]. Retrieved from https://martin-thoma.com/nlp-reuters.

14. 20 Newsgroups. [n.d.]. Retrieved from http://qwone.com/jason/20Newsgroups/.
15. Wang, F., Wang, Z., Li, Z., & Wen, J. R. (2014, November). Concept-based short text classification and ranking. In Proceedings of the 23rd ACM International Conference on Conference on Information and Knowledge Management (pp. 1069-1078).
16. Greene, D., & Cunningham, P. (2006, June). Practical solutions to the problem of diagonal dominance in kernel document clustering. In Proceedings of the 23rd international conference on Machine learning (pp.377-384).
17. Das, A. S., Datar, M., Garg, A., & Rajaram, S. (2007, May). Google news personalization: scalable online collaborative filtering. In Proceedings of the 16th international conference on World Wide Web (pp. 271-280).
18. CNN-News18 Headlines with Descriptions. [n.d.]. Retrieved from https://www.kaggle.com/datasets/mauryansshivam/cnn-news18-headlines-with-descriptions.
19. News Articles Dataset from Indian Express. [n.d.]. Retrieved from https://www.kaggle.com/datasets/pulkitkomal/news-article-data-set-from- indian-express.
20. NEWS SUMMARY. [n.d.]. Retrieved from https://www.kaggle.com/datasets/sujalbhagathansda/news-summary.
21. Conneau, Alexis, et al. "Unsupervised cross-lingual representation learning at scale." arXiv preprint arXiv:1911.02116 (2019).
22. Amari, Shun-ichi. "Backpropagation and stochastic gradient descent method." Neurocomputing 5.4-5 (1993): 185-196.
23. Kingma, Diederik P., and Jimmy Ba. "Adam: A method for stochastic optimization." arXiv preprint arXiv:1412.6980 (2014).